

Design a Car Rental System

Let's design a Car Rental System

We'll cover the following:

- System Requirements
- Use Case Diagram
- Class Diagram
- Activity Diagrams

A Car Rental System is a software built to handle the renting of automobiles for a short period of time, generally ranging from a few hours to a few weeks. A car rental system often has numerous local branches (to allow its user to return a vehicle to a different location), and is primarily located near airports or busy city areas.



Figure 1: Car Rental System

1 System Requirements

We will focus on the following set of requirements while designing our Car Rental System:

1. The system will support the renting of different automobiles like cars, trucks, SUVs, vans, and motorcycles.
2. Each vehicle should be added with a unique barcode and other details, including a parking stall number, which helps to locate the vehicle.
3. The system should be able to retrieve information like which member took a particular vehicle or what vehicles have been rented out by a specific member.
4. The system should collect a late fee for vehicles returned after the due date.
5. Members should be able to search the vehicle inventory and reserve any available vehicle.
6. The system should be able to send notifications whenever the reservation is approaching the pick-up date, as well as when the vehicle is nearing the due date or has not been returned within the due date.
7. The system will be able to read barcodes from vehicles.
8. Members should be able to cancel their reservations.
9. The system should maintain a vehicle log to track all events related to the vehicles.
10. Members can add rental insurance to their reservation.
11. Members can rent additional equipment, like navigation, child seat, ski rack, etc.
12. Members can add additional services to their reservation, such as roadside assistance, additional driver, wifi, etc.

2 Use Case Diagram

We have four main actors in our system:

- **Receptionist:** Mainly responsible for adding and modifying vehicles and workers. Receptionists can also reserve vehicles.
- **Member:** All members can search the catalog, as well as reserve, pick up, and return a vehicle.
- **System:** Mainly responsible for sending notifications about overdue vehicles, canceled reservations, etc.
- **Worker:** Mainly responsible for taking care of returned vehicles and updating the vehicle log.

Here are the top use cases of the Car Rental System:

- **Add/Remove/Edit vehicle:** To add, remove, or modify a vehicle.
- **Search catalog:** To search for vehicles by type and availability.
- **Register new account/Cancel membership:** To add a new member or cancel an existing membership.
- **Reserve vehicle:** To reserve a vehicle.
- **Check-out vehicle:** To rent a vehicle.
- **Return a vehicle:** To return a vehicle that was checked out to a member.
- **Add equipment:** To add equipment to a reservation like navigation, child seat, etc.
- **Update car log:** To add or update a car log entry, such as refueling, cleaning, damage, etc.

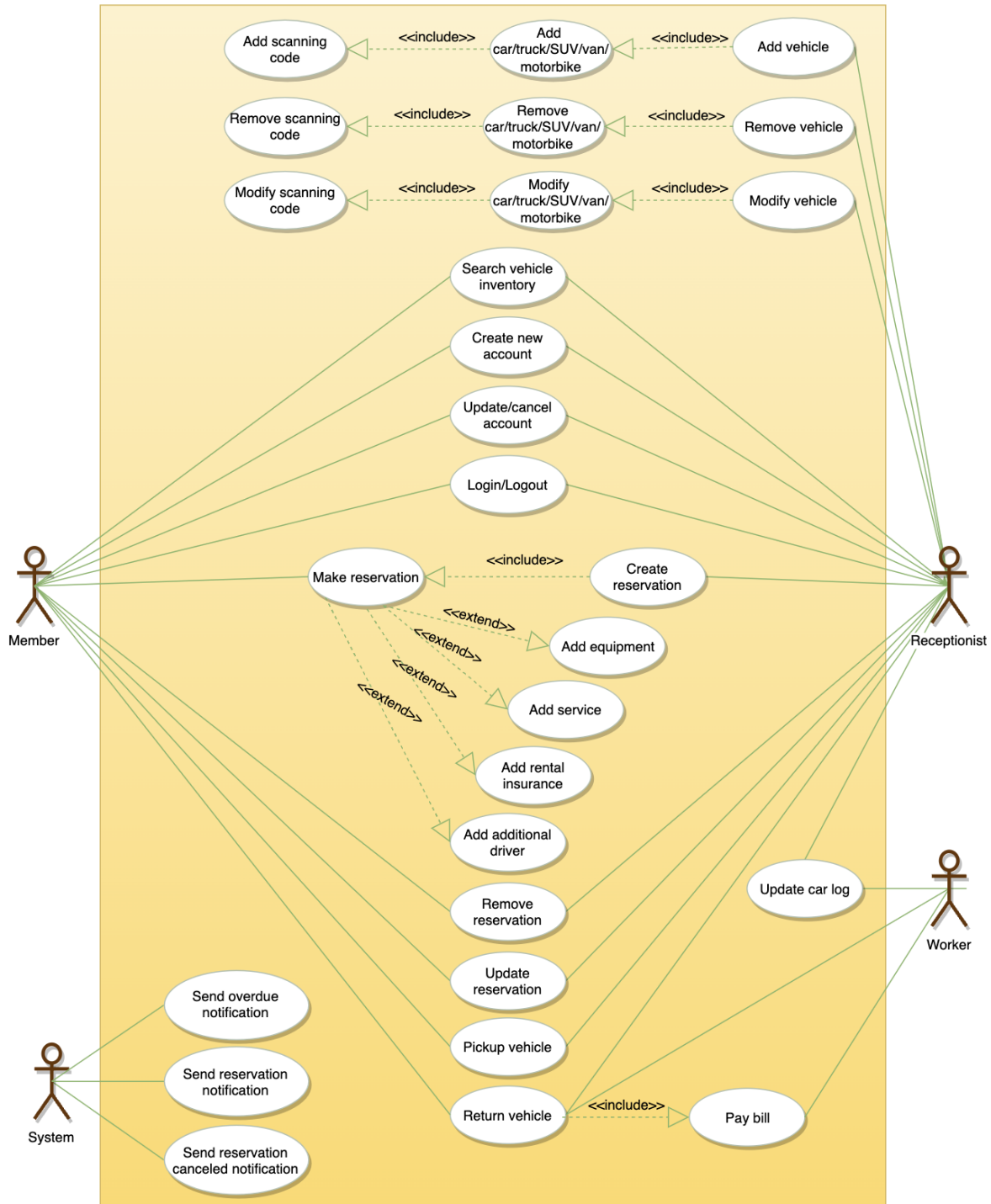


Figure 2: Use Case Diagram for Car Rental System

3 Class Diagram

Here are the main classes of our Car Rental System:

- **CarRentalSystem:** The main part of the organization for which this software has been designed.

- **CarRentalLocation:** The car rental system will have multiple locations, each location will have attributes like 'Name' to distinguish it from other locations and 'Address' which defines the address of the rental location.
- **Vehicle:** The basic building block of the system. Every vehicle will have a barcode, license plate number, passenger capacity, model, make, mileage, etc. Vehicles can be of multiple types, like car, truck, SUV, etc.
- **Account:** Mainly, we will have two types of accounts in the system: a general member and a receptionist. Another account can be for the worker who takes care of returned vehicles.
- **VehicleReservation:** This class will be responsible for managing reservations for a vehicle.
- **Notification:** Responsible for sending notifications to members.
- **VehicleLog:** Keeps track of all the events related to a vehicle.
- **RentalInsurance:** Stores details about the various rental insurances that members can add to their reservation.
- **Equipment:** Stores details about the various types of equipment that members can add to their reservation.
- **Service:** Stores details about the various types of services that members can add to their reservation, such as additional drivers, roadside assistance, etc.
- **Bill:** Contains different bill items for every charge for the reservation.

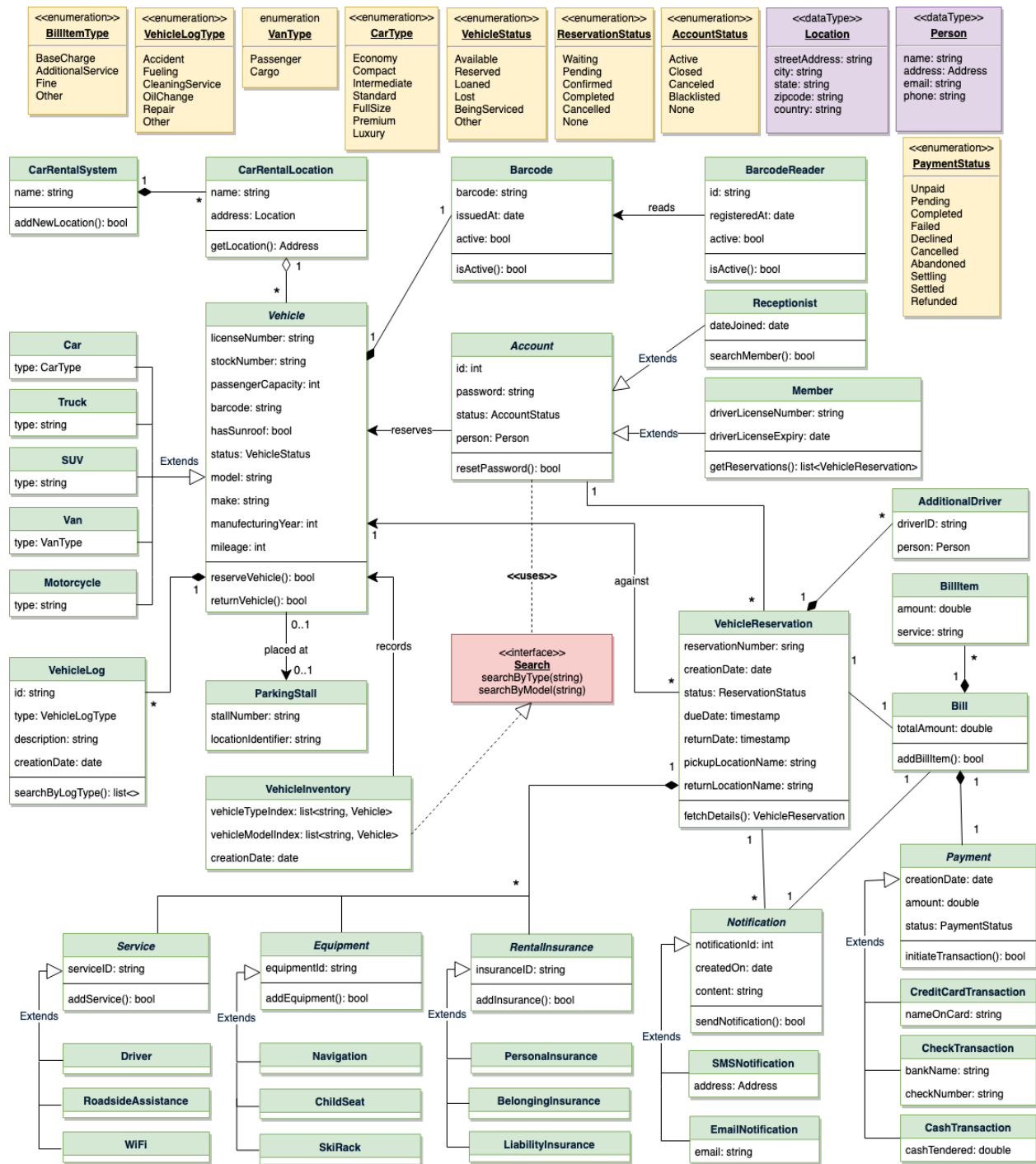


Figure 3: Class Diagram for Car Rental System

4 Activity Diagrams

Pick up a vehicle: Any member can perform this activity. Here are the steps to pick up a vehicle:

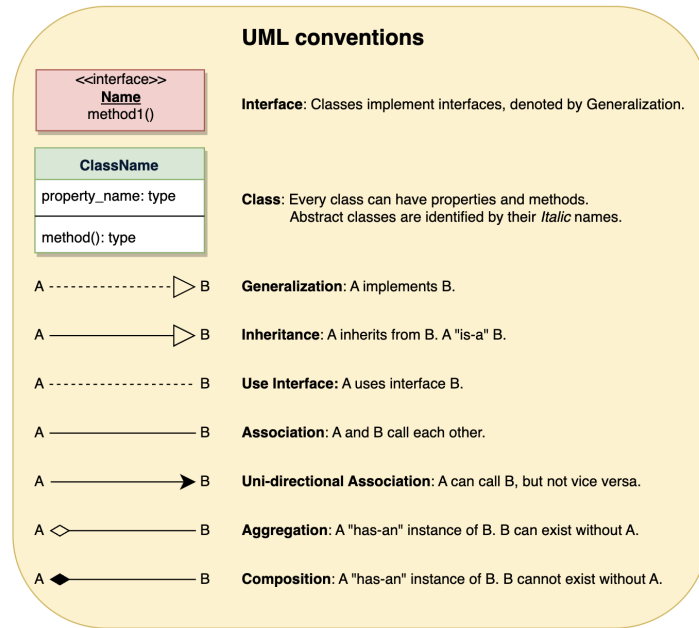


Figure 4: UML for Car Rental System

Return a vehicle: Any worker can perform this activity. While returning a vehicle, the system must collect a late fee from the member if the return date is after the due date. Here are the steps for returning a vehicle:

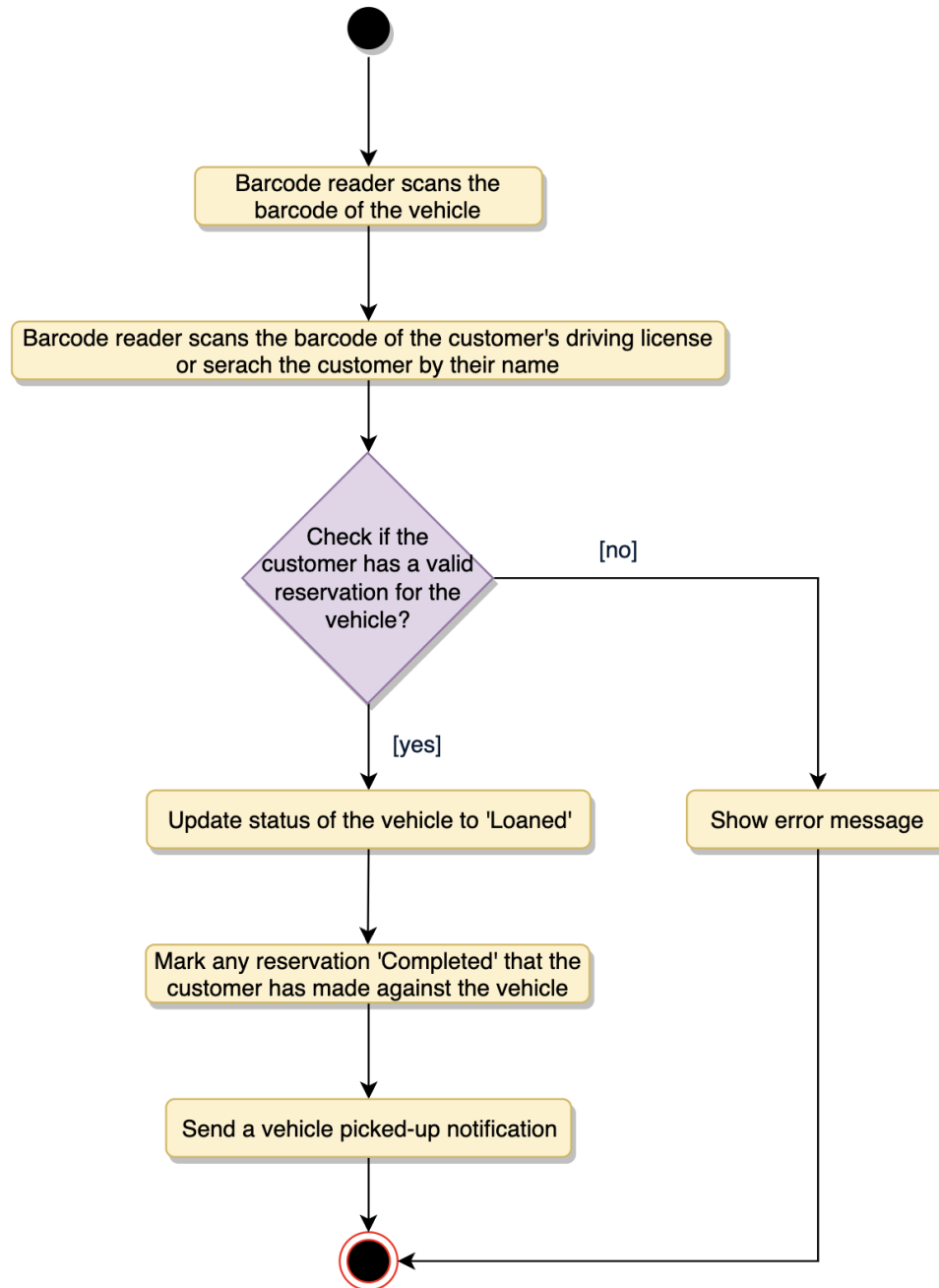


Figure 5: Activity Diagram for Car Rental System Pick Up

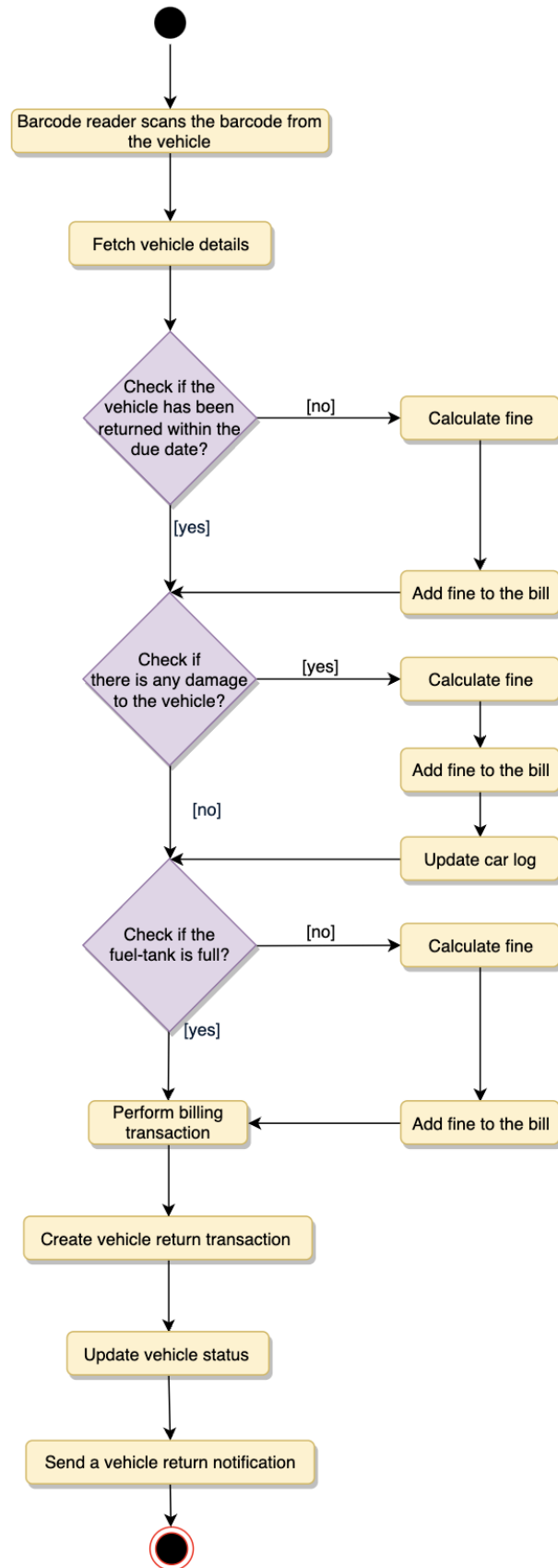


Figure 6: Activity Diagram for Car Rental System Return